



School-Wide Network

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PROJECT ON

School-Wide Network

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School-Wide Network

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CERTIFICATE

This is to certify that this report titled “School-Wide Network” embodies the original work done by Aldi Syarif Alhakim, Alezuna M. Nadhif Pohan, Muhamad Ihsan Rasyad. Project in partial fulfillment of their course requirement at CEP-CCIT FTUI.

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M. Riza Iqbal Latief, S.T.

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Finally, we appreciate all group members for their cooperation and teamwork in completing the School-Wide Network project.

BACKGROUND

Internet access has become an essential part of the learning process in schools. Students use the internet to access educational materials, complete assignments, conduct research, and support communication activities. However, unmanaged network infrastructure can lead to bandwidth congestion, unstable connections, security vulnerabilities, and inefficient resource distribution.

To address these issues, a School-Wide Network is designed to provide a structured and centralized network infrastructure that connects laboratories, classrooms, and administrative areas. The system integrates wired (LAN) and wireless connections under a single router-based management system to ensure stable and reliable connectivity across the school environment.

By implementing this network design, the school can improve performance, enhance security through network segmentation, manage bandwidth efficiently, and create a more organized and scalable network system to support academic and administrative activities.

SYSTEM ANALYSIS

The School-Wide Network is designed to provide centralized and structured network connectivity across the entire school environment. The system supports both wired and wireless connections to ensure stable and efficient communication for academic and administrative activities.

In this design, laboratory computers are directly connected to the server room using wired LAN connections to ensure high stability, low latency, and reliable performance during practical sessions. Meanwhile, classrooms and teacher areas access the network through wireless access points distributed across the school to provide flexible and wide coverage.

The network uses a centralized router connected to an Internet Service Provider (ISP) as the main gateway. The router manages IP addressing, internet distribution, bandwidth control, and network security. Switches are used to distribute connections from the server room to laboratory devices and access points.

By implementing this architecture, the system improves network stability in laboratory environments, provides flexible wireless access for other areas, enhances centralized management, and ensures efficient bandwidth utilization throughout the school.

SECTIONS	NETWORK ADDRESS	SUBNETMASK	AVAILABLE HOST
LAB	192.168.40.0	255.255.255.0	253
GURU	192.168.30.0	255.255.255.0	253
SISWA	192.168.20.0	255.255.255.0	253
MURID	192.168.10.0	255.255.255.0	253
MANAGEMENT	192.168.99.0	255.255.255.0	253

NETWORK DEVICES

LAB

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point2	-	=	-
Computer	PC	192.168.40.6 (DHCP) 192.168.40.9 (DHCP) 192.168.40.5 (DHCP) 192.168.40.7 (DHCP) 192.168.40.8 (DHCP) 192.168.40.12 (DHCP) 192.168.40.3 (DHCP) 192.168.40.10 (DHCP) 192.168.40.13 (DHCP) 192.168.40.11 (DHCP)	192.168.40.1	Wireless
Web Cam IOT	Webcam	192.168.40.4 (DHCP)	192.168.40.1	Wireless

TEACHER ROOM

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point1	-	-	-
Computer	PC	192.168.30.12 (DHCP) 192.168.30.2 (DHCP) 192.168.20.11 (DHCP)	192.168.30.1	
Web Cam IOT	Webcam	192.168.30.5 (DHCP) 192.168.30.9 (DHCP)	192.168.30.1	Wireless
SmartPhone	Smartphone	192.168.30.3 (DHCP) 192.168.30.6 (DHCP) 192.168.30.13 (DHCP)	192.168.30.1	
Laptop PT	Laptop	192.168.30.7 (DHCP) 192.168.30.15 (DHCP) 192.168.30.14 (DHCP) 192.168.30.4 (DHCP) 192.168.30.8 (DHCP) 192.168.30.10 (DHCP)	192.168.30.1	

CLASSROOM LEFT

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point6	-	-	-
Smartphone-PT	Smartphone	192.168.10.10 (DHCP) 192.168.10.14 (DHCP) 192.168.10.2 (DHCP) 192.168.10.20 (DHCP) 192.168.10.17 (DHCP) 192.168.10.9 (DHCP) 192.168.10.5 (DHCP) 192.168.10.13 (DHCP) 192.168.10.21 (DHCP) 192.168.10.18 (DHCP) 192.168.10.11 (DHCP) 192.168.10.22 (DHCP) 192.168.10.23 (DHCP) 192.168.10.24 (DHCP) 192.168.10.4 (DHCP) 192.168.10.14 (DHCP) 192.168.10.15 (DHCP) 192.168.10.12 (DHCP) 192.168.10.25 (DHCP) 192.168.10.27 (DHCP) 192.168.10.19 (DHCP)	192.168.10.1	

		192.168.10.6 (DHCP) 192.168.10.7 (DHCP) 192.168.10.3 (DHCP)		
Web Cam IOT	Web Cam	192.168.10.8 (DHCP) 192.168.10.16 (DHCP) 192.168.10.6 (DHCP)	192.168.10.1	Wireless

CANTEEN

Devices	Devices Name	IP Address	Gateway	Features
Web Cam IOT	Webcam	192.168.20.16 (DHCP)	192.168.20.1	Wireless
SmartPhone	Smartphone	192.168.20.20 (DHCP) 192.168.20.14 (DHCP) 192.168.20.36 (DHCP) 192.168.20.19 (DHCP) 192.168.20.22 (DHCP) 192.168.20.24 (DHCP)	192.168.20.1	

CLASSROOM RIGHT

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point6	-	-	-
Smartphone-PT	Smartphone	192.168.20.40 (DHCP) 192.168.20.49 (DHCP) 192.168.20.39 (DHCP)	192.168.20.1	

		192.168.20.42 (DHCP) 192.168.20.28 (DHCP) 192.168.20.27 (DHCP) 192.168.20.13 (DHCP) 192.168.20.48 (DHCP) 192.168.20.4 (DHCP) 192.168.20.6 (DHCP) 192.168.20.5 (DHCP) 192.168.20.10 (DHCP) 192.168.20.8 (DHCP) 192.168.20.9 (DHCP) 192.168.20.11 (DHCP) 192.168.20.15 (DHCP) 192.168.20.18 (DHCP) 192.168.20.23 (DHCP) 192.168.20.28 (DHCP) 192.168.20.29 (DHCP) 192.168.20.30 (DHCP) 192.168.20.32 (DHCP) 192.168.20.33 (DHCP) 192.168.20.35 (DHCP)		
WebCam-IOT		192.168.20.27 (DHCP) 192.168.20.34 (DHCP) 192.168.20.25 (DHCP) 192.168.20.31	192.168.20.1 (DHCP)	wireless

		(DHCP) 192.168.20.58 (DHCP)		
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Office and Server Room

Devices	Devices Name	IP Address	Gateway	Features
Server-PT	server0	192.168.99.3	192.168.99.1	-
Switch	Multilayer Switch	-	-	-
Cloud PT	Cloud	-	-	-
WLC-2504	Wireless LAN C.	192.168.99.2	-	-
PC-PT	Computer	192.168.40.15 (DHCP)	192.168.40.1	
WebCam-IOT	Webcam	192.168.40.2 (DHCP)	192.168.40.1	-Wireless

Configuration IP Address (Server 1 & 2)

Server 1 (DHCP + DNS)
Command Prompt

“nmcli connection add type ethernet ifname ens224 con-name hostonly \”

“ipv4.addresses 192.168.10.2/24 ipv4.method manual”

“nmcli connection up hostonly”

```
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group de
fault qlen 1000
    link/ether 00:0c:29:4e:30:b0 brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    altname enx000c294e30b0
    inet 192.168.10.2/24 brd 192.168.10.255 scope global noprefixroute ens224
        valid_lft forever preferred_lft forever
    inet6 fe80::c8fd:4fa9:c67d:273/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@localhost:/home/redhat#
```

Server 2 (Web Server)
Command Prompt

“nmcli connection add type ethernet ifname ens224 con-name hostonly \”

“ipv4.addresses 192.168.10.3/24 ipv4.method manual”

“nmcli connection up hostonly”

```
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen
1000
    link/ether 00:0c:29:cd:2a:be brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    altname enx000c29cd2abe
    inet 192.168.10.3/24 brd 192.168.10.255 scope global noprefixroute ens224
        valid_lft forever preferred_lft forever
    inet6 fe80::1a80:c4da:ea12:11f0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@localhost:/home/redhat#
```

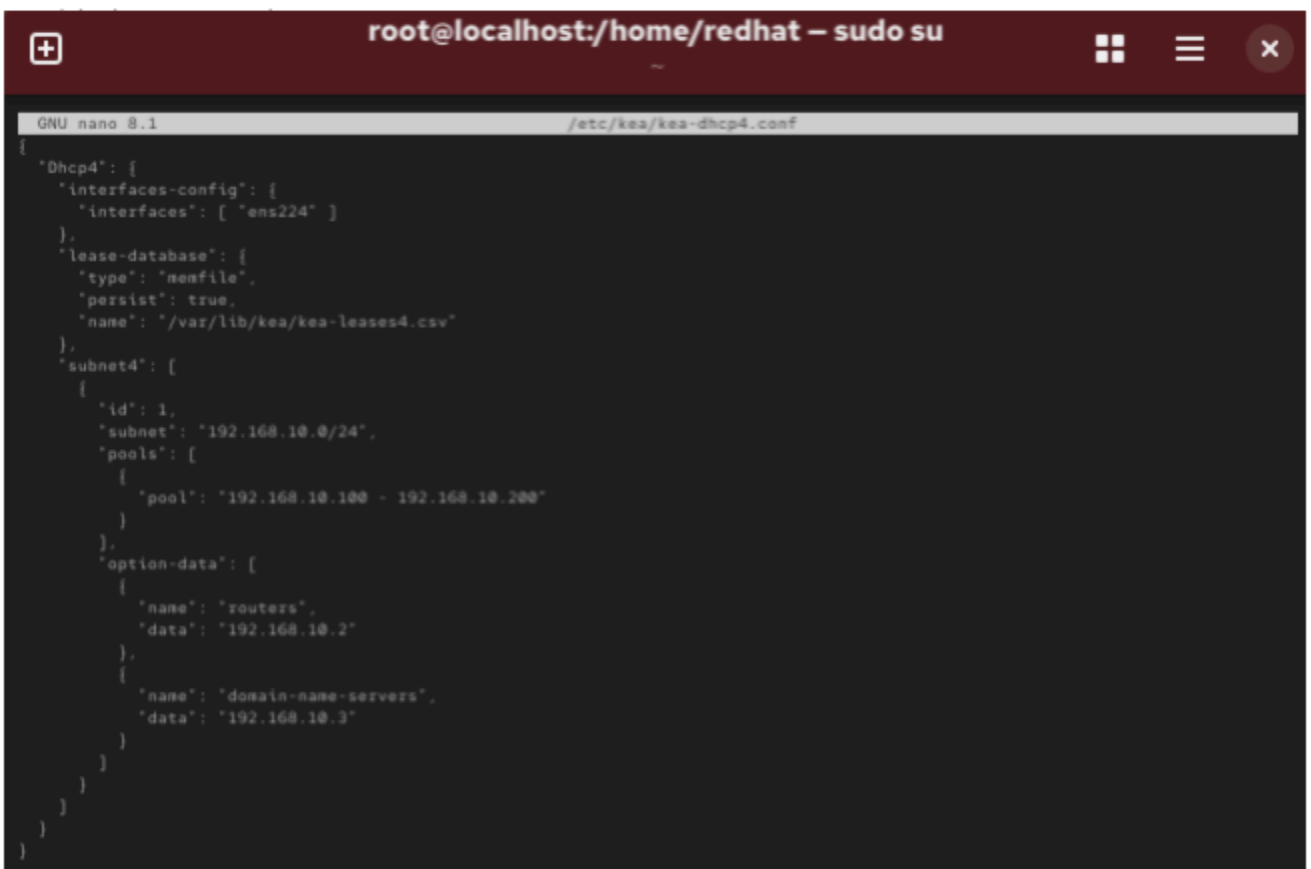
DHCP Server & Configuration (kea DHCP)

first download kea dhcp in redhat version 10

“dnf install kea -y”

After that, do the configuration according to the IP we want.

“nano /etc/kea/kea-dhcp4.conf”



The screenshot shows a terminal window with a dark red title bar. The title bar contains the text "root@localhost:/home/redhat – sudo su" and three icons: a plus sign, a hamburger menu, and a close button. The terminal content shows the GNU nano 8.1 editor editing the file /etc/kea/kea-dhcp4.conf. The configuration is as follows:

```
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "ens224" ]
    },
    "lease-database": {
      "type": "newfile",
      "persist": true,
      "name": "/var/lib/kea/kea-leases4.csv"
    },
    "subnet4": [
      {
        "id": 1,
        "subnet": "192.168.10.0/24",
        "pools": [
          {
            "pool": "192.168.10.100 - 192.168.10.200"
          }
        ],
        "option-data": [
          {
            "name": "routers",
            "data": "192.168.10.2"
          },
          {
            "name": "domain-name-servers",
            "data": "192.168.10.3"
          }
        ]
      }
    ]
  }
}
```

DHCP Server & Configuration

(kea DHCP)

After configuration, we can start running DHCP.

“systemctl enable kea-dhcp4”

“systemctl start kea-dhcp4”

After that, we can check if our DHCP is running successfully.

“systemctl status kea-dhcp4

```
root@localhost:/home/redhat# systemctl status kea-dhcp4
● kea-dhcp4.service - Kea DHCPv4 Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset>
   Active: active (running) since Sun 2026-03-29 13:32:35 WIB; 3min 7s ago
 Invocation: a36ca7ab50c542d49e7e272bfd42dd88
    Docs: man:kea-dhcp4(8)
  Main PID: 4515 (kea-dhcp4)
   Status: "Dispatching packets..."
    Tasks: 7 (limit: 10144)
  Memory: 3.4M (peak: 6.2M)
     CPU: 234ms
    CGroup: /system.slice/kea-dhcp4.service
           └─4515 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf
```

DNS Server (BIND)

After completing dhcp we start installing dns
“dnf install bind bind-utils -y”

Once finished we can start configuring
“nano /etc/named.conf”

```
root@localhost:/home/redhat – sudo su
GNU nano 8.1 /etc/named.conf
include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";

zone "sekolah.local" IN {
    type master;
    file "sekolah.local.db";
};

^G Bantuan      ^O Tulis        ^F Cari         ^K Potong       ^T Jalankan     ^C Lokasi
^X Keluar      ^R Baca         ^\ Ganti        ^U Tempel      ^J Ratakan     ^/ Ke Baris
```

after configuration we create a zone file
“nano /var/named/[sekolah.local.db](#)”

```
root@localhost:/home/redhat – sudo su
GNU nano 8.1 /var/named/sekolah.local.db
$TTL 86400
@ IN SOA ns.sekolah.local. root.sekolah.local. (
    2026032901 ; Serial
    3600      ; Refresh
    1800      ; Retry
    604800    ; Expire
    86400     ; Minimum TTL

@ IN NS ns.sekolah.local.
ns IN A 192.168.10.3
@ IN A 192.168.10.3
www IN A 192.168.10.3
```

DNS Server (BIND)

Once finished we can start setting permissions
"chown root:named /var/named/sekolah.local.db"
"chmod 640 /var/named/sekolah.local.db"

continue running after setting permission
"systemctl enable named"
"systemctl start named"

Lastly, firewall
"firewall-cmd --permanent --add-service=dns"
"firewall-cmd --reload"

WEB SERVER (NGINX + PHP)

after dns we go to the web server

“dnf install nginx php php-fpm php-mysqlnd -y --nogpgcheck”

After installing, run it

“systemctl enable nginx php-fpm”

“systemctl start nginx php-fpm”

After running it, we set up the website

“nano /etc/nginx/conf.d/default.conf”



```
root@localhost:/home/redhat – sudo su
GNU nano 8.1 /etc/nginx/conf.d/default.conf
server {
    listen      80;
    server_name localhost;

    root        /usr/share/nginx/html;
    index        index.php index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include      fastcgi_params;
        fastcgi_pass  unix:/run/php-fpm/www.sock;
        fastcgi_index index.php;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
    }
}
```

WEB SERVER (NGINX + PHP)

Simple Login

First, create a file that will become our website.
move to the file that we will fill in the file. (cd)
After that, fill in the file, here there are 4 files (touch)

```
root@localhost:/home/redhat# cd /usr/share/nginx/html
root@localhost:/usr/share/nginx/html# touch index.php proses.php dashboard.php logout.php
```

After that, we configure it in order, starting from the left.
"nano index.php"



```
root@localhost:/usr/share/nginx/html - sudo su
GNU nano 8.1 index.php
<?php
session_start();

// kalau sudah login, langsung ke dashboard
if (isset($_SESSION['user'])) {
    header("Location: dashboard.php");
    exit();
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Login Sekolah</title>
</head>
<body>
    <h2>Login</h2>

    <form method="POST" action="proses.php">
        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <button type="submit">Login</button>
    </form>
</body>
</html>
```

WEB SERVER (NGINX + PHP)

Simple Login

After our index we go to process
“nano proses.php”

```
root@localhost:/usr/share/nginx/html – su
GNU nano 2.8.1 proses.php
<?php
session_start();

// validasi input
if (!isset($_POST['username']) || !isset($_POST['password'])) {
    die("Akses tidak valid!");
}

$conn = new mysqli("127.0.0.1", "webuser", "123", "sekolah");

if ($conn->connect_error) {
    die("Koneksi gagal: " . $conn->connect_error);
}

$username = $_POST['username'];
$password = $_POST['password'];

$sql = "SELECT * FROM users WHERE username='$username' AND password='$password'";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    $_SESSION['user'] = $username;
    header("Location: dashboard.php");
    exit();
} else {
    echo "Login gagal! <br><a href='index.php'>Coba lagi</a>";
}

$conn->close();
?>
```

WEB SERVER (NGINX + PHP)

Simple Login

After the process we go to the dashboard
"nano dashboard.php"

```
root@localhost:/usr/share/r
GNU nano 8.1 dashboard
<?php
session_start();

// kalau belum login
if (!isset($_SESSION['user'])) {
    die("Akses ditolak! Silakan login dulu.");
}

<!DOCTYPE html>
<html>
<head>
    <title>Dashboard</title>
</head>
<body>
    <h2>Dashboard</h2>

    <p>Selamat datang, <b><?php echo $_SESSION['user']; ?></b></p>

    <a href="logout.php">Logout</a>
</body>
</html>
```

finally we log out
"nano logout.php"

```
root@
GNU nano 8.1
<?php
session_start();
session_destroy();

header("Location: index.php");
exit();
?>
```

DATABASE (MariaDB)

after our web to the database

“dnf install mariadb-server -y --nogpgcheck”

After we install it, we start

“systemctl enable mariadb”

“systemctl start mariadb”

setup

“mysql -u root -p”

fill

“

```
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 29
Server version: 10.11.15-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE sekolah;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [sekolah]> INSERT INTO users (username, password) VALUES ('aldi', '123');
Query OK, 1 row affected (0,001 sec)

MariaDB [sekolah]> INSERT INTO users (username, password) VALUES ('rasyad', '123');
Query OK, 1 row affected (0,000 sec)

MariaDB [sekolah]> INSERT INTO users (username, password) VALUES ('nadhif', '123');
Query OK, 1 row affected (0,002 sec)

MariaDB [sekolah]> SELECT * FROM users;
+----+-----+-----+
| id | username | password |
+----+-----+-----+
| 1 | admin    | 123      |
| 2 | aldi     | 123      |
| 3 | rasyad   | 123      |
| 4 | nadhif   | 123      |
+----+-----+-----+
```

WEB + DATABASE CONNECT

Setting

"nano /usr/share/nginx/html/index.php"

fill



The screenshot shows a terminal window with a red title bar that reads "root@localhost:/home/redhat – sudo su". Inside the terminal, the GNU nano 8.1 text editor is open, editing the file "/usr/share/nginx/html/index.php". The code in the editor is a PHP script that connects to a MySQL database, queries a table named 'users', and displays the results in an HTML table. The code is as follows:

```
<?php
$conn = new mysqli("127.0.0.1", "webuser", "123", "sekolah");

if ($conn->connect_error) {
    die("Koneksi gagal: " . $conn->connect_error);
}

$sql = "SELECT * FROM users";
$result = $conn->query($sql);

echo "<h2>Data User</h2>";

if ($result->num_rows > 0) {
    while($row = $result->fetch_assoc()) {
        echo "ID: " . $row["id"] . " - Username: " . $row["username"] . "<br>";
    }
} else {
    echo "Tidak ada data";
}

$conn->close();
?>
```

Fix SELinux

setsebool -P httpd_can_network_connect_db 1

Client

ip request to dhcp server
"nmcli connection down ens160"
"nmcli connection up ens160"

cek
"ip a & cat /etc/resolv.conf" (DHCP)

```
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group de
fault qlen 1000
    link/ether 00:0c:29:41:97:c0 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    altname enx000c294197c0
    inet 192.168.10.100/24 brd 192.168.10.255 scope global dynamic noprefixroute
    ens160
        valid_lft 5845sec preferred_lft 5845sec
    inet6 fe80::20c:29ff:fe41:97c0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@localhost:/home/redhat# cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.10.3
root@localhost:/home/redhat#
```

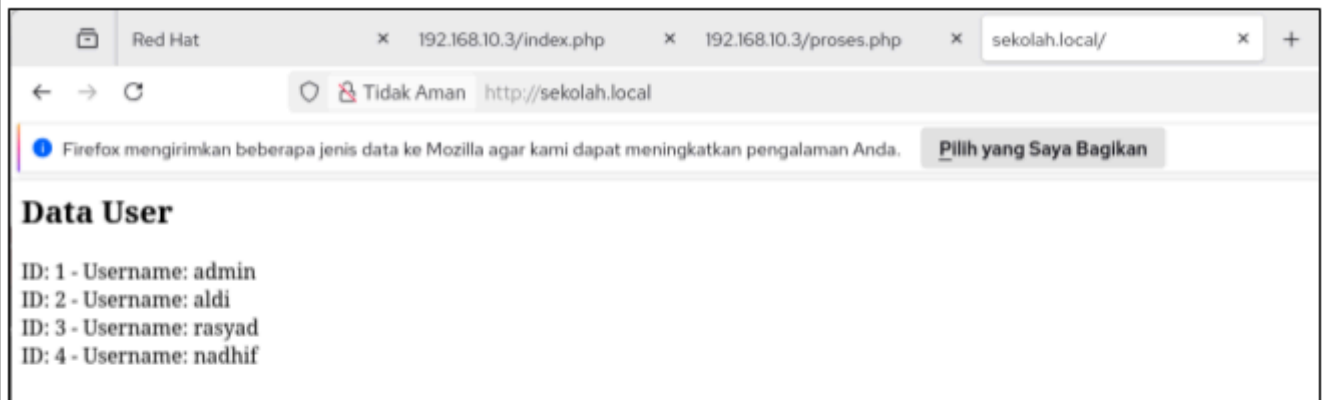
"nslookup sekolah.local" (DNS)

```
root@localhost:/home/redhat# nslookup sekolah.local
Server:          192.168.10.3
Address:         192.168.10.3#53

Name:   sekolah.local
Address: 192.168.10.3
root@localhost:/home/redhat#
```

Client

“<http://sekolah.local>” (data base + web + DNS)



“<http://192.168.10.3>” (data base + web + ip)

